



The well-being of software engineers: a systematic literature review and a theory

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Abstract

For decades, software engineering research and practice has focused primarily on technological and process-related factors. Today, there is a growing interest in organizational, social, and psychological factors, including well-being. Organizational studies show that well-being contributes to work outcomes, including creativity, performance, and productivity. But despite its importance, the predictors and outcomes of software engineers' well-being as a multidimensional construct to date are under-researched. This paper reports on the multidimensional well-being of professional software engineers and generates insights for the future research in this area. 44 quantitative survey studies published between 2000 and 2023 were selected and synthesized both quantitatively and qualitatively through a systematic literature review. The results of the review were further analyzed to construct a quantitatively-testable theory, detailing the predictors and outcomes of well-being in software engineering organizations. The total number of research participants included in the selected studies is 16,086 software engineering professionals from at least 42 countries. The literature review identified various measures, constructs, and indicators of well-being, as well as its predictors and outcomes. The theory, based on cumulative results of carefully selected quantitative studies, is an attempt to “correct the record” by establishing well-being in software engineering as a meta-construct of hedonic, eudaimonic, and integrated or hedaimonic qualities predicted by different individual, team and organizational factors and impacting the functioning of software engineers and their organizations. The review highlighted the under-researched aspects of well-being in software engineering and confirmed the need for more advanced quantitative studies. We hope that the theory will benefit researchers in conducting future studies and practitioners in developing nuanced and science-based interventions for improving software engineers' well-being.

Keywords Software engineering · Well-being · Positive psychology · Theory-building · Systematic literature review · SLR

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1 Introduction

The goal of positive psychology (i.e., the science of well-being) is to understand and increase the flourishing of individuals, communities, and societies (Seligman and Csikszentmihalyi 2000; Seligman 2011). The relationship between positive attitudes and work performance is considered the "Holy Grail" of organizational studies (Weiss and Cropanzano 1996). Recent systematic literature reviews show that well-being is associated with better work outcomes, such as performance, productivity, creativity, citizenship, and customer satisfaction (Salas-Vallina et al. 2018; García-Buades et al. 2019). In software engineering research, an increased number of studies are investigating well-being using innovative methods, such as sentiment analysis (Mäntylä et al. 2018a, b). However, software engineers' well-being as a multidimensional construct is still under-researched (Lenberg et al. 2015; Sánchez-Gordón and Colomo-Palacios 2019), especially due to a lack of quantitative psychological studies.

Many researchers nowadays agree that well-being is a multidimensional construct that can include physical, emotional, cognitive, and social aspects of personal, professional, and collective functioning (Dodge 2012; Lomas et al. 2014; Linton et al. 2016). However, to the best of our knowledge, there are no comprehensive multidimensional models or theories of well-being that describe and explain what makes software engineers flourish in their personal and professional lives (Singh and Junnarkar 2016). In fact, many studies seem to reduce the software engineers' well-being to the absence of negative affective states, such as distress and burnout (Maudgalya et al. 2006; Chilton et al. 2010; Kuriakose et al. 2019; Rezvani and Khosravi 2019; Mellblom et al. 2019; Heshimova et al. 2020; Kuuttila et al. 2020; Zaidman and Cohen 2020), or the presence of positive affective states, such as positive affect or motivation (Graziotin et al. 2015a, b, 2017, 2018; Ke and Zhang 2018). Therefore, the bias towards unidimensional affective/motivational states is a serious conceptual issue.

In this paper, we report our results from conducting a systematic literature review on well-being of software engineers and synthesize the evidence from existing studies to arrive at a theory that explains the relationships between the predictors (antecedents) and outcomes (consequences) of software engineers' well-being. Our research is driven by the following research questions:

- RQ1. What are the known constructs, indicators, and measures of well-being relevant for software engineers?
- RQ2. What are the known predictors of well-being of software engineers?
- RQ3. What are the known outcomes of well-being of software engineers?

We have addressed these research questions through a systematic literature review and formulated a theory of well-being of software engineers based on carefully selected quantitative studies. The purpose of the developed theory is to provide a rationale for further advanced psychological studies of well-being in software engineering, and to provide actionable advice for leaders to increase the well-being of their employees.

The rest of the paper is structured as follows. Section 2 outlines related research and existing models of well-being in general and in software engineering in particular. Section 3 details the steps of the systematic literature review, including the search strategy, study selection, data extraction, quality assessment, the methodology behind the data synthesis and theory development, and the deviations from the protocol. Section 4 reports the

results of the literature review, which are further used to create the theory of well-being of software engineers. The results, the theory, and practical recommendations are discussed in Section 5. Finally, Section 6 concludes the paper with key contributions to practice and research, limitations, and future research directions. Additional information, including the review protocol, search terms, and links to datasets containing the search and analysis results are available upon request.

2 Background

2.1 Definition of Well-Being

According to Pollard and Lee (2003), well-being is "a complex, multi-faceted construct that has continued to elude researchers' attempts to define and measure" (p. 60). In positive psychology, definitions and measures of general well-being are grounded in at least three traditions (Dodge et al. 2012; Ng and Fisher 2013; Huta 2015; Cookie 2016):

- hedonic well-being marked by higher levels of life satisfaction and positive emotion and lower levels of negative emotion;
- eudaimonic well-being marked by higher levels of meaning and self-determination;
- integrated well-being marked by constructs sharing both hedonic and eudaimonic qualities, such as higher levels of flow and engagement.

Hedonic well-being is one of the oldest and most extensively studied traditions, rooted in ancient Greek philosophy. According to Aristippus, a Greek philosopher and the founder of the Cyrenaic school of hedonist philosophy (fourth century BCE), pleasure is the only good and pain is the only evil in life; therefore, the goal of life is to seek pleasure and avoid pain (Ng and Fisher 2013; Huta 2015). Other well-known hedonist thinkers who emphasized the importance of pleasure over pain include Thomas Hobbes (psychological egoism), Jeremy Bentham (utilitarianism), and Sigmund Freud (pleasure principle). In psychological terms, hedonic well-being can be defined as the presence of positive affect or pleasant emotional arousal (e.g., joy, enthusiasm, and determination) and the absence of negative affect or unpleasant emotional arousal (e.g., sadness, distress, and irritation) (Watson et al. 1988; Weiss and Cropanzano 1996). However, contemporary conceptualizations of hedonic well-being (rebranded as "subjective well-being" or "SWB") usually include an emotional component marked by high positive affect and low negative affect and a cognitive component marked by high life satisfaction (Diener 1984; Diener et al. 1985; Pavot and Diener 2008).

Eudaimonic well-being is also rooted in Greek philosophy. In *Nicomachean Ethics*, Aristotle wrote that true happiness is pursued by living a "good life" marked by reason and virtue rather than hedonistic pursuits (Huta 2015). In modern terms, eudaimonia (rebranded as "psychological well-being" or "PWB") is experienced while fulfilling psychological needs for autonomy, competence, environmental mastery, growth, self-acceptance, positive relationships, and purpose (Ryff 1989; Ryff and Keyes 1995; Ryff and Singer 2006). Similarly, self-determination theory argues that when people fulfill their needs for autonomy,

competence, and relatedness or relationships, they are more intrinsically motivated to pursue eudaimonic activities (Deci and Ryan 2000; Ryan et al. 2006).

Integrated well-being is a relatively modern attempt to bring hedonic and eudaimonic traditions together (Tulip et al. 2020). It is mostly associated with the PERMA model that defines well-being as a composite factor of positive emotions, engagement, positive relationships, meaning, and accomplishment or achievement (Seligman 2011, 2018). In our view, integrated well-being is also closely linked to experiences of flow and engagement. According to Mihaly Csikszentmihalyi (1975, 2008), the developer of flow theory, we experience flow when we immerse and lose our selves in tasks a little bit beyond our current skills and abilities (marker of eudaimonic well-being). For this reason, the state can be highly rewarding and enjoyable (marker of hedonic well-being). At work, we experience flow or absorption—one of the markers of engagement—when we are fully focused and engrossed in our work (Schaufeli et al. 2002; Bakker et al. 2008).

In the clinical literature, we can find another tradition that conceptualizes well-being as a form of health (Jahoda 1958; Engel 1997). For example, the World Health Organization¹ defines health as “a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity”. In the two-continua model of mental health continuum, well-being is marked by the presence of flourishing (i.e., high mental health) and absence of languishing (i.e., low mental health or mental illness) (Keyes 2002, 2005). We consider the clinical tradition also as a form of integrated well-being that emphasizes positive states (marker of hedonic well-being) and positive functioning (marker of eudaimonic well-being). Furthermore, Linton et al. (2016) reviewed around 100 measures of health and well-being and grouped them into: physical (i.e., bodily functioning), mental (i.e., cognitive and emotional quality of life), social (i.e., relationships and social interactions), and spiritual well-being (i.e., meaning and self-transcendence). Linton’s research concluded that well-being measures strongly resemble general health measures. However, it should be noted that the paper focused mainly on measures used in healthcare rather than organizational settings and did not explore existing conceptual or theoretical differences between health and well-being.

With respect to organizational settings, the last two decades marked an increased interest in employees’ job satisfaction, happiness, and other positive states (García-Buades et al. 2019; Bhatia and Mohsin 2020). The growing interest is associated with the radical shift from machine bureaucracy to increased adhocracy in the organizational contexts (Mintzberg 1984; Lee and Edmondson 2017) and the emergence of new trends in organizational research, such as positive organizational behavior (Luthans 2002), positive organizational scholarship (Cameron et al. 2003), and positive organizational psychology (Donaldson and Ko 2010).

According to the review by Salas-Vallina et al. (2018), the happiness of knowledge-intensive workers is indicated by various constructs, including commitment (i.e., connection with an organization), engagement (i.e., optimal functioning at work), job satisfaction (i.e., positive appraisal of one’s work), psychological capital (i.e., optimism, efficacy, resilience, and hope), and well-being (i.e., hedonic and eudaimonic). Furthermore, the review also shows that the most prominent theoretical framework of happiness at work is the Job Demands-Resources Model. The model holds that happiness is the result of the appraised

¹ World Health Organization. (1948, April 7). Preamble to the Constitution of the World Health Organization (WHO). <https://www.who.int/about/who-we-are/constitution>

balance between job demands (e.g., role ambiguity, time pressure) and job resources (e.g., autonomy, feedback) (Bakker et al. 2005, 2007). The review also shows that happiness at work indicates the absence of ill-being, i.e., lack of stress and burnout, whereas happiness constitutes the presence of well-being marked by an emotional component (i.e., positive affect) and a cognitive component (i.e., life satisfaction).

2.2 Health and Well-Being in Software Engineering

For many decades, software engineering research and practice has focused primarily on technological and process-related factors with very little attention to organizational, social, or psychological factors (Lenberg et al. 2015). However, the shift from linear plan-driven waterfall approaches to iterative agile methods that rely on self-organization instead of "top-down" direction (Galindo Bello 2018) also fueled the interest in psychological factors. Management philosophy in agile environments values individuals and interactions more than processes and tools (Fowler and Highsmith 2001), contributing to team performance, project outcomes, software quality, and a more balanced workload (Venkatesh et al. 2020). Successful use of agile methods highly depends on individual and team behaviors and characteristics, such as openness for innovation (Vishnubhotla et al. 2020) and interpersonal skills (Boehm and Turner 2003; Hummel et al. 2013), which ultimately changes the professional profile of software engineers.

Software engineering as a profession unites various specialists, both technical (e.g., programmers and testers) and not so technical (e.g., analysts and product managers).² Based on the data of various systematic literature reviews, software engineers are believed to have distinct socioeconomic, motivational, emotional, and temperamental characteristics. Socioeconomically, they are considered intelligent, well-educated, and high-paid (Maudgalaya et al. 2006). Motivationally, they are driven by technical challenges, problem-solving, and benefiting society (Beecham et al. 2008; França et al. 2011). Emotionally, they are more prone to negative emotions (fear, sadness, and anger) than positive ones (Sánchez-Gordón et al. 2019). Temperamentally, they tend to be more introverted than extroverted and rely more on their thoughts, sensations, and judgments than on feelings, intuitions, and perceptions (Cruz et al. 2015). Based on the above, software engineers' interests, motivation, and temperament might be at odds with the social nature of modern software development, which can be detrimental to their mental health and well-being—an important field of study.

Mental health and well-being are important aspects of software engineers' lived experience. Related studies demonstrate that software engineers work in a dynamic environment marked by tight deadlines, high workload, task interdependence, and uncertainty (Kunda 1992; Kameo 2017; Zaidman and Cohen 2020), which are especially relevant for agile contexts (Venkatesh et al. 2020). Therefore, it is not surprising that software engineers perceive elevated levels of stress and exhaustion (Thong and Yap 2000; Windeler et al. 2017; Suárez and Vizcaíno 2023). Second, increasing focus on social skills and teamwork has been found to be sources of potential exhaustion and job dissatisfaction, especially when team members lack interpersonal skills (Venkatesh et al. 2020), are less agreeable and less extroverted (Acuña et al. 2015; Rodrigues and Rebelo 2020). On the contrary, emotionally

² Stack Overflow. (2023). Stack Overflow developer survey. Available online: <https://insights.stackoverflow.com/survey/2023>

intelligent software engineers were found to experience more job satisfaction and less stress (Rodrigues and Rebelo 2020). Third, time pressure, stress, and exhaustion have been associated with a number of negative outcomes, including programming errors, high maintenance costs, reduced performance, increased turnover (Windeler et al. 2017; Venkatesh et al. 2020), and unethical behavior, such as shortcut taking and code plagiarism (Sojer et al. 2014; Kuuttila et al. 2020). Finally, industry research indicates that many software engineers experience psychological distress. Based on highly representative cross-cultural data collected by Stack Overflow (2018, 2019, 2020, 2021, 2022),³ approximately 8.5% of professional software engineers are increasingly suffering from mood and anxiety disorders, which is twice as high as for the general population (Vos et al. 2017). Similarly, software engineers are found to sleep less and be more prone to sleep disturbances, such as mild and severe insomnia (Zadeh and Begum 2010).

One may believe that software engineering profession is associated with particularly negative mental health. Yet, proneness to psychological suffering does not necessarily mean the absence of well-being. Research suggests that individuals can experience complex states of well-being simultaneously comprising of both positive and negative affect (Berrios 2017a, b, 2018). Contrary to Stack Overflow results, Graziotin et al. (2017, 2018) demonstrated that professional software engineers experience slightly more positive affect than other studied groups, such as students and workers from other areas. There are several possible reasons why software engineers may feel happier than other populations. First, software engineers have a high potential to experience positive states like engagement (Alsunki et al 2020) and flow (Murphy 2014; Pratt et al. 2016), leading to higher levels of performance (Graziotin et al. 2014a) and productivity (Graziotin et al. 2014b) and lower levels of turnover (Ghapanchi and Aurum 2011). Finally, research shows that time pressure and stress are not always detrimental to well-being, as they can be perceived as a positive challenge rather than a hindrance (Chong et al. 2011).

The abovementioned findings contribute to the debate about the uniqueness of software engineering as a profession. The negative professional characteristics are worrisome and deserve further attention. Related research also suggests that the different aspects of software engineers' well-being, due to their complex interplay, cannot be studied in isolation. Therefore, there is a need for a more holistic understanding of software engineers' well-being and related factors.

3 Methodology

In this paper, we report our findings from a systematic literature review that aimed at the theoretical conceptualization of the well-being in software engineering. Although there are other literature reviews of software engineers' well-being, these focus either on unidimensional states of well-being, such as emotions (Sánchez-Gordón and Colomo-Palacios 2019) and motivation (Beecham et al. 2008; França et al. 2011), or negative markers of

³ The average sample size of survey participants who responded to questions on mental health and neurodiversity is 13,915 participants from up to 186 countries (see "Stack Overflow" in our Online appendix). Reports are available online for separate years: 2018 (<https://insights.stackoverflow.com/survey/2018>); 2019 (<https://insights.stackoverflow.com/survey/2019>); 2020 (<https://insights.stackoverflow.com/survey/2020>); 2021 (<https://insights.stackoverflow.com/survey/2021>); 2022 (<https://insights.stackoverflow.com/survey/2022>); 2023 (<https://insights.stackoverflow.com/survey/2023>).

well-being, such as stress (Maudgalya et al. 2006; Suárez and Vizcaíno 2023), time pressure (Kuutila et al. 2020), and cognitive load (Gonçales et al. 2021). In our review, we summarized evidence from quantitative studies and developed a state-of-the-art vocabulary and roadmap with clear research gaps and directions for future research.

The literature review adopted the guidelines by Kitchenham (2007, 2015) accepted as the common guidelines for executing systematic literature reviews in software engineering and used in other related reviews of software engineering studies (Beecham et al. 2008; Cruz et al. 2015; Sánchez-Gordón and Colomo-Palacios 2019). The details of each step of the literature review are documented in the following sub-sections. Please refer to our Online appendix⁴ for more information on our methodology and research data (available upon request).

3.1 Search Strategy

We performed an automated search in six digital libraries, two indexing services, and one professional network (see Table 1). Contrary to the recommended practice (Kitchenham 2007, 2015), we did not perform a manual search, such as forward snowballing (i.e., search via citations) or backward snowballing (i.e., search via reference lists) to retain the repeatability of the search results and avoid the bias that could arise from selective use of references and citations. After all, researchers are prone to confirmation bias, i.e., the tendency to verify rather than falsify one's claims (Hergovich et al. 2010; Lilienfeld 2017; Schumm 2021).

Following various guidelines for performing systematic literature reviews in software engineering (Kitchenham 2007, 2015; Wohlin et al 2013; Stapic et al. 2016), we selected ACM, IEEE, ScienceDirect, and Scopus as databases that cover the most established journals and conference proceedings. We added PubMed and EBSCO, since they have appeared in relevant systematic reviews of well-being studies outside the software engineering literature (Linton et al. 2016; Batz-Barbarich et al. 2018; Gonot-Schoupinsky and Garip 2018; García-Buades et al 2019). Finally, we added ResearchGate to reduce publication bias, as it includes conference papers and papers in print that might not be indexed in digital libraries and indexing services. However, due to the limited search engine of ResearchGate, Google Search was used as an intermediary search engine to obtain results, e.g., *researchgate.net/publication/happiness* “*software developers*” (refer to section “Search Strings” in our Online appendix).

Search terms related to well-being are based on popular constructs and indicators of hedonic, eudaimonic, and integrated well-being (see Table 2). The diverse conceptualizations are discussed in many reviews of well-being (Dodge 2012; Ng and Fisher 2013; Huta 2015; Cookie 2016). To narrow down the search results to the software engineering field, we added terms related to software engineering, including programmers, software engineers, software development, software engineering, and software engineers, as suggested in other related reviews. We considered adding the term “programming”, but after piloting the search in ACM, Wiley, and EBSCO, we realized that it results in too many irrelevant papers, related to educational programs or genetic programming. Finally, the pilot search

⁴ <https://docs.google.com/spreadsheets/d/1TzAACVK8uGLtAOFoTskZigtS0JQ0N51IvzWCdhxChFE/edit?gid=622967878&gid=622967878>

Table 1 Selected search engines

Search engine	Type	Abbreviation
ACM Digital Library	Digital library	ACM
EBSCOHost	Indexing service	EBSCO
• Academic Search Complete • APA PsycArticles • APA PsycInfo • Business Source Complete • CINAHL Plus with Full Text • Communication & Mass Media Complete • Education Research Complete • Library, Information Science & Technology Abstracts		
ProQuest	Digital library	ProQuest
PubMed	Digital library	PubMed
IEEE Xplore	Digital library	IEEE
Researchgate via Google Search	Professional network	Researchgate
ScienceDirect	Digital library	ScienceDirect
Scopus	Indexing service	Scopus
Wiley Online Library	Digital library	Wiley

demonstrated that search engines had recognized *well-being* (standard spelling) as two separate words; therefore, *wellbeing* was included instead.

It should be noted that the list of terms used in this systematic literature review is not exhaustive. For example, in terms of well-being, we did not include specific terms like *physical well-being* and *collective well-being*, assuming that the general term *wellbeing* will pick up relevant studies. Also, we excluded terms associated with the absence of well-being, such as negative affect and stress, because negative states do not directly indicate well-being. Similarly, we omitted many synonyms to software engineers (e.g., software professional or IT professional) and terms that describe other software development roles (e.g., testing or product management). This was done to control the length of the search

Table 2 Well-being-related terms used in this study

Tradition	Model/measure	Associated search terms
Hedonic	Subjective well-being (Diener 1984; Diener et al. 1985)	Happiness, positive affect, satisfaction, wellbeing
Eudaimonic	Six-factor model of psychological well-being (Ryff 1989; Ryff and Keyes 1995; Ryff and Singer 2006)	Autonomy, environmental mastery, flourishing, personal growth, positive relations, purpose in life, self-acceptance, wellbeing
	Self-determination theory (Deci and Ryan 2000; Ryan et al. 2006)	Autonomy, competence, relatedness, wellbeing
Integrated	Flow (Csikszentmihalyi 1975, 2008)	Flow, happiness
	Mental health continuum (Keyes 2002, 2005)	Flourishing, health, mental health, wellbeing, wellness
	PERMA (Seligman 2011, 2018)	Achievements, accomplishments, engagement, flourishing, meaning, positive emotion (-s), positive relationships, wellbeing

strings and thus increase the replicability of the search process (Kitchenham 2007, 2015). Also, increasing the number of terms increases the number of irrelevant studies, which is a recognized as a common issue in the reviews in software engineering studies (Kitchenham 2007). Therefore, replicability was favored over completeness.

Based on the search terms, a search string (see Table 3) was devised to perform an automated search in three iterations. First, a small segment of the string [(*“software engineering”* OR *“software engineers”*) AND (*“wellbeing”* OR *“well-being”*)], was piloted in EBSCO, ScienceDirect, Scopus, ResearchGate, and JSTOR. Unfortunately, JSTOR did not return any studies of software engineers, so we removed it from the list of databases (see section “Pilot Search” on Online appendix). Second, we used the complete string to search the years 2000–2021 (the concluding year when the study was conducted for a master’s thesis). Because not all search engines allowed for the use of a lengthy string, we split the search string into smaller strings to search on ScienceDirect and ResearchGate via Google Search, and adapted the syntax of the string to search on IEEE (see section “Search Strings” on Online appendix). Finally, in order to include the latest papers during the preparation of this article, we extended the search with the years 2021–2023, using the same search engines, except for Researchgate (to reduce the number of duplicates) and ProQuest (we lost access to the database and were unable to regain it). Overall, the automatic search identified 44,301 records (see Fig. 1).

3.2 Study Selection

The studies were selected based on justified inclusion and exclusion criteria (see Table 4). First, we have included all studies published in academic venues, such as scientific journal articles and conference papers, written in English and published between 2000 and 2023. We chose to start with 2000s, when the focus shifted from process and tools to individuals and interactions (Fowler and Highsmith 2001) and when positive organizational behavior (Luthans 2002) and positive organizational scholarship (Cameron et al. 2003) were introduced. Second, we focused only on quantitative studies in order to perform a fair comparison among the studies. Also, quantitative methods are the most common in well-being research (Helwig et al. 2020). Third, we excluded studies on such aspects as cognitive load, stress, and time pressure, since these affective states alone do not constitute well-being. Finally, we excluded studies focusing on students of software engineering, unemployed software engineers, and workers not related to software engineering.

To ensure the quality of the study selection process, the authors compared their inclusion and exclusion decisions by going through a sample of 100 articles obtained through ACM Digital Library using the search strategy (see Section 3.1). The agreement between the authors was 100 percent: both authors excluded 99 articles; one article was initially

Table 3 Default search string used in this study

(*“accomplishments”* OR *“achievements”* OR *“autonomy”* OR *“competence”* OR *“engagement”* OR *“environmental mastery”* OR *“flourishing”* OR *“flow”* OR *“happiness”* OR *“health”* OR *“meaning”* OR *“personal growth”* OR *“positive affect”* OR *“positive emotion”* OR *“positive emotions”* OR *“positive relations”* OR *“positive relationships”* OR *“purpose in life”* OR *“relatedness”* OR *“satisfaction”* OR *“self-acceptance”* OR *“self-determination”* OR *“wellbeing”* OR *“wellness”*) AND (*“programmers”* OR *“software developers”* OR *“software development”* OR *“software engineering”* OR *“software engineers”*)

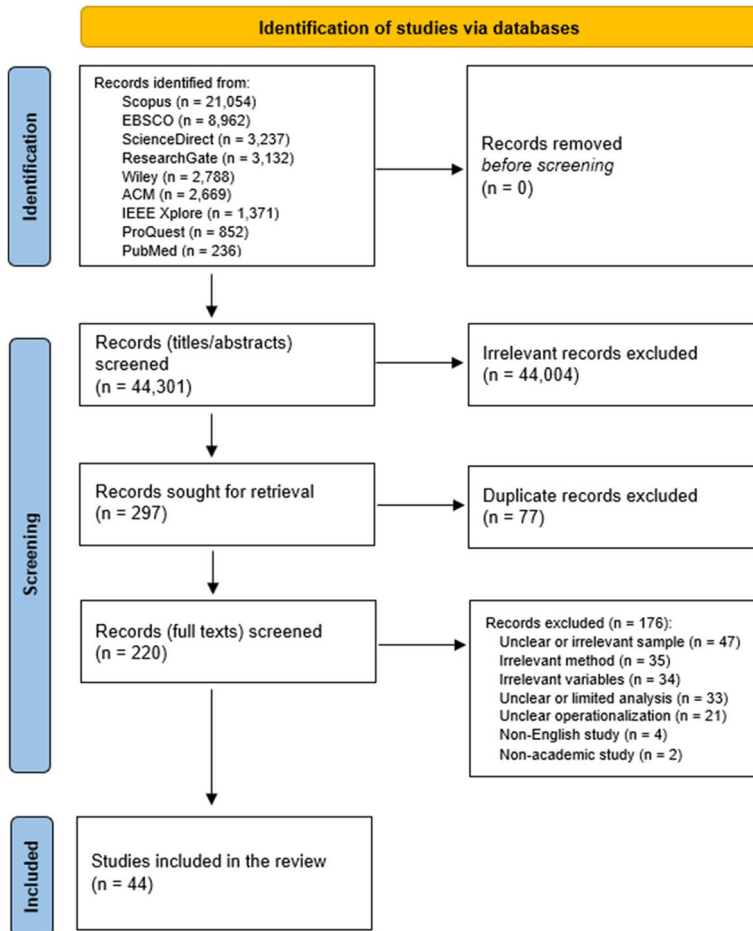


Fig. 1 PRISMA flow diagram, based on McKenzie et al. 2021)

Table 4 Inclusion and exclusion criteria and their rationale

Inclusion criterion	Exclusion criterion
Published in a scientific journal or conference proceeding	Dissertations, white papers, technical reports, etc
Published in English language	Published in a language other than English
Published 2000–2023	Published before Jan 2000 or after 2023 Dec
Quantitative study measuring well-being	Qualitative or theoretical studies
Studied variables constitutes or indicates well-being (see Table 2)	Well-being was not studied
The sample includes mostly software engineering professionals (e.g., programmers, testers, analysts, designers, project managers)	The sample includes mostly students, unemployed software engineers, or workers not related to software engineering or IT
The study meets the quality criteria for rigor (relevant variables are clearly operationalized) and credibility (regression coefficients and their significance levels are presented)	The study does not meet the quality criteria for rigor (relevant variables are clearly operationalized) and credibility (regression coefficients and their significance levels are not presented)

considered but excluded by both authors due to a lack of important statistical information. The rest of the selection process was conducted, documented, and justified by the first author (see Online appendix for relevant information).

After screening the titles (and abstracts when the relevance was unclear from the title), 44,004 records were excluded, 297 records were sought for retrieval, 77 duplicates were excluded, and 220 full-text records were assessed for eligibility. The significantly high number of irrelevant studies is natural in the automatic search of software development studies, especially when many terms are included in the search (Kitchenham 2007). After screening the full-text information, 44 studies were included in the review, while 176 were excluded. The study was excluded if it used non-correlational or non-experimental methods (e.g., case or field study, information extraction study); studied irrelevant variables (i.e., not constructs or indicators of well-being); surveyed irrelevant samples (e.g., students, hobbyists); lacked clear operationalization of relevant variables; lacked clear analytical information (e.g., correlation/regression coefficients and their significance values); demonstrated features of a non-academic study; was non-English.

3.3 Data Extraction

Data extraction was guided by the research questions and performed manually. Data extracted includes the following:

- General information on the publication (IDs; title; date; authors; institutions; publication types; publication venues);
- General information on the methodology (research participants: sample size, sample location, gender, professional role; research methods);
- Information on the study of well-being: research measures applied to study well-being; constructs/indicators of well-being; predictors and outcomes of well-being; results of statistical tests and relationships (correlations and regressions).

Data was extracted by the first author, who is a qualified psychologist with a background in quantitative survey studies, to ensure the accuracy of the construct understanding. The extracted data is available in our Online appendix.

3.4 Quality Assessment

The quality assessment (QA) was performed by the first author as an additional exclusion step to screen out studies of low quality. The QA criteria were adapted from Yang et al. (2021) who synthesized the most common criteria in a recent tertiary study. The QA included various stages (i.e., design, execution, analysis, and conclusion) and characteristics (i.e., rationality, rigor, credibility, and contribution). For this study, the stages and characteristics were combined into four final categories, while the statements were transformed into questions (see Table 5).

Realizing that simple yes/no answers can be unreliable (Kitchenham 2007), the studies were evaluated on a scale from 1 to 4, where 1 – strongly disagree, 2 – disagree, 3 – agree, and 4 – strongly agree. The quality scores were calculated as follows: the sum of evaluations divided by the total sum of evaluations (i.e., 56) and multiplied by 100 (see section “Quality” in the Online appendix for the quality assessment results). According

Table 5 The quality assessment criteria used in this study

Category	Criterion/question
Rationale	RA.1 Is the context clearly described? RA.2 Are the reasons for conducting the study clearly described? RA.3 Are the aims clearly described? RA.4 Are the hypotheses clearly described?
Rigor	RI.1 Is the data collection method clearly described and justified? RI.2 Is the sample clearly described and justified? RI.3 Are the measures clearly described and justified? RI.4 Is the data analysis method clearly described and justified?
Credibility	CR.1 Are results (including positive and negative) clearly described? CR.2 Are the results supported by other studies? CR.3 Is the relationship between the researcher and participants clearly described? CR.4 Are limitations, biases, or threats to the validity and reliability clearly described?
Contribution	CO.1 Is further research clearly described? CO.2 Is the value to research or practice clearly described?

to this methodology, a study can have a very low quality (1–24), low to average quality (25–49), average to good quality (50–74), and very good quality (75–100). The quality scores can explain the study results' differences (Kitchenham 2007). QA scores are as follows: total average = 75, total median = 77, lowest = 46, highest = 95. Also, we provide the average scores for separate quality categories: rationale = 84, rigor = 68, credibility = 72, and contribution = 80.

Please note that, initially, we included all of the selected studies that received, on average, at least 50 points (medium to good quality). However, upon further analysis, we noticed that three of the selected studies with the lowest quality ratings (i.e., WB01 = 52, WB02 = 50, and WB03 = 46) were correlational only and did not explore the direction of the statistically significant effects or relationships between variables, which prevented us from using the results of these studies in our theoretical model. For this reason, we decided to include these low-quality studies in the descriptive synthesis (see Sects. 4.1 and 4.2, but exclude them from the qualitative synthesis (see Section 4.3).

3.5 Data Synthesis and Theory Development

The general information extracted from the studies was synthesized descriptively in figures and tables, whereas the main findings of the selected studies were synthesized in two steps.

In the first step, we refined and categorized the indicators, predictors, and outcomes of well-being using semantic analysis (i.e., looking at semantic similarities and differences between variables) and deductive reasoning (i.e., moving from general theory to specific conclusions). To ease the navigation through a large number of variables, we divided them into three levels:

1. variable category (the lowest level represented by the actual variables analyzed in the selected studies),

2. main category (the mid-level represented by semantically and theoretically related variables), and
3. core category (the highest level represented by semantically and theoretically related main categories).

In terms of the indicators of well-being, we grouped variables, such as *job satisfaction* and *work satisfaction*, into the main category of *satisfaction*, because both indicators measure how respondents are satisfied with certain aspects of their jobs and professional lives. Furthermore, if established theories of well-being combined certain variables into broader constructs, these theories would guide the clustering process. For example, factors like *life satisfaction* and *positive affect* would constitute *subjective* or *hedonic well-being*, whereas experiences of *autonomy*, *competence*, *relatedness*, and *meaning* would constitute *psychological* or *eudaimonic well-being* (see Section 2.1). To identify whether certain constructs shared either more similarities or differences, we would also consult the semantics of the respective survey items and questions that were adopted to collect data on a particular indicator of well-being. As a result, we arrived at a two-level categorization with three core categories: *hedonic*, *eudaimonic*, and *hedaimonic* or *integrated well-being*. The same process was organized to homogenize the predictors and outcomes of well-being.

In terms of the predictors of well-being, variables that describe personality traits, such as *extroversion* or *neuroticism*, would constitute the main category of *personality predictors*, while variables that describe career orientations, such as *managerial* and *technical*, would constitute the main category of *career motivation predictors*. Consequently, *personality* and *career motivation* would be further subsumed into the core category of individual factors, because both main categories describe individual traits and preferences of software engineering professionals that they bring to their organizations. Similarly, main categories, such as *job predictors*, *team predictors*, and *workplace predictors*, would constitute the core category of *organizational predictors*, which are circumstances that are “brought onto” software engineering professionals by their teams and organizations.

In the second step, we identified and synthesized the relationships between indicators of well-being and their respective predictors and outcomes using a form of inductive reasoning, that is, moving from specific observations to general theory. In this case, our observations were guided by the statistical significance of the relationships between well-being and its predictors and outcomes. Due to the differences in the measures of well-being and research participants, we were unable to perform a meta-analysis, which prevented us from calculating common effect sizes and providing more validity to the theory. Nevertheless, the results of the review still empowered us to develop a testable theory and its visualization (see Section 4.3) using the guidelines and recommendations proposed by Sjøberg et al. (2008).

The statistically significant relationships between the variables were used to formulate theoretical propositions. The goal of these propositions is to establish which factors predict which constructs of well-being, and which constructs of well-being predict which outcomes. As such, our theory is a type III theory, as defined by Gregor (2006). In contrast to theories that guide analysis and description (Type I) and theories that guide understanding (Type II), predictive theories (Type III) are expected to predict what will happen if certain conditions hold with an approximate or probabilistic degree of certainty (Gregor 2006). The theoretical propositions can be used to test the theory through further empirical studies, whereas primary studies can be consulted to construct explanations and justifications.

The general information of the included studies, such as the geographic representation of roles of research participants constitute the scope of the theory.

The data synthesis was performed by the first author who is knowledgeable in psychology and verified by the second author knowledgeable in software engineering. In case of question, the primary studies were revisited and discussed. The theory development was performed in close collaboration between both authors. The first author led the description of the theory core concepts and implications, while the second author assisted in formalizing the description of the theory, according to the guidelines proposed by Sjøberg et al. (2008) and Stol and Fitzgerald (2015).

The results of the data synthesis and theory development are presented in Section 4. To ensure transparency and usability, we kept the diagrammatic representation of our theory on two levels: operational (Fig. 4 and Fig. 5) and conceptual (Fig. 6). The operational level provides references to the primary studies and the indicators of well-being analyzed in the selected studies, while the conceptual level provides a holistic, higher-level abstraction of the theoretical relationships between the constructs.

3.6 Deviations From the Protocol

The original protocol (see the “Protocol” in our Online appendix) was reviewed by two reviewers at the School of Psychology, University of East London, and piloted by the first author against the software engineering research literature. Based on the feedback and pilot search, some changes were made. First, the review type was changed from an integrative review to a systematic literature review of software development studies. Also, the number of associated systematic literature reviews relevant for the background of the study was increased. Then, the mixed search strategy was replaced with an automated search. Finally, the rationale, aims, questions, search strategy and terms, inclusion/exclusion criteria, data extraction, quality assessment, and publication plans were expanded or redefined.

4 Results

4.1 Overview of the Included Quantitative Studies of Well-Being

In this section, we provide an overview of general characteristics of the quantitative studies of well-being included in the literature review. The selected studies⁵ (WB01-WB44) are listed in Appendix 1.

The number of quantitative studies dedicated to well-being of software engineers divided into 4-year intervals shows a growing trend, which suggests an increasing interest in software engineers’ well-being (see Fig. 2).

The majority of selected studies, i.e., about 89% ($n=39$), are journal articles, whereas about 11% ($n=5$) are conference papers. Detailed analysis shows that journal articles demonstrate higher average and median quality ($x=77$, $\bar{x}=79$) than conference papers ($x=62$, $\bar{x}=64$) (see section “Paper types” in our Online appendix). Finally, an analysis of excluded papers indicates that more conference papers were excluded from the review than journal

⁵ Please note that if a selected study followed a mixed-method approach, we would present and analyze only relevant quantitative results.

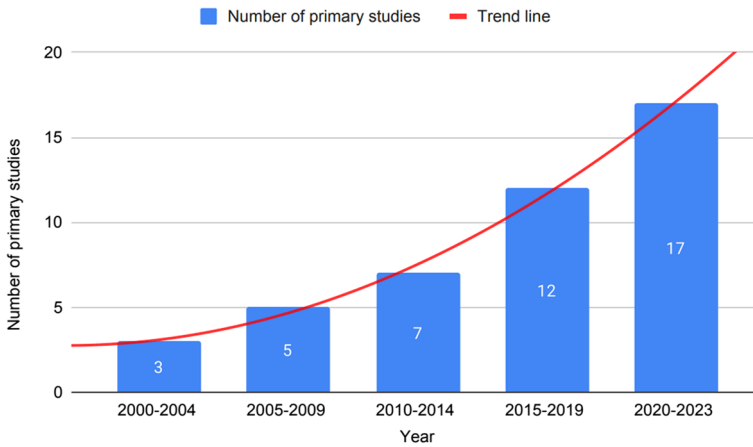


Fig. 2 Increase in number of primary studies by intervals

articles due to quality issues associated with unclear operationalization of relevant variables as well as unclear or limited analysis.

More than a half of the selected studies ($n=23$) were published in software engineering and technology-related venues, such as Empirical Software Engineering ($n=3$), Information & Management ($n=3$), Journal of Management Information Systems ($n=2$), Journal of Systems and Software ($n=2$), IEEE Transactions on Software Engineering ($n=2$), and International Workshop on Cooperative and Human Aspects of Software Engineering ($n=2$). However, studies of software engineers' well-being were also published in prominent journals related to management and psychology, including the Journal of Applied Psychology ($n=3$).

The selected studies include insights from the total of 16,086 software engineers ($n=44$). The participants whose location is disclosed ($n=12,225$ or 76%) originate from 42 countries ($n=32$) (see Fig. 3). Further analysis of the participants suggests that the average percentage of women in the total sample is 27% ($n=32$). Finally, the most studied software engineering populations focus on software development ($n=44$), product and project management ($n=12$), analysis ($n=10$), as well as testing and quality assurance or QA ($n=9$) (see Table 6).

4.2 Overview of the Approaches and Measures to Study Well-Being

Here, we report on how well-being was approached in the included studies as a research construct and a measure.

The majority of studies were cross-sectional ($n=40$), whereas the remaining were longitudinal ($n=3$) and quasi-experimental ($n=1$) (see Table 7). The most common statistical methods were structural equation modeling ($n=18$) and regression analysis ($n=17$), followed by multilevel modeling ($n=5$), and correlation analysis ($n=4$) (see Table 8).

Most studies operationalized well-being by using existing items and measures ($n=25$), whereas a smaller number of studies either adapted existing items and measures according their needs ($n=10$), referred to the literature to create their own items and

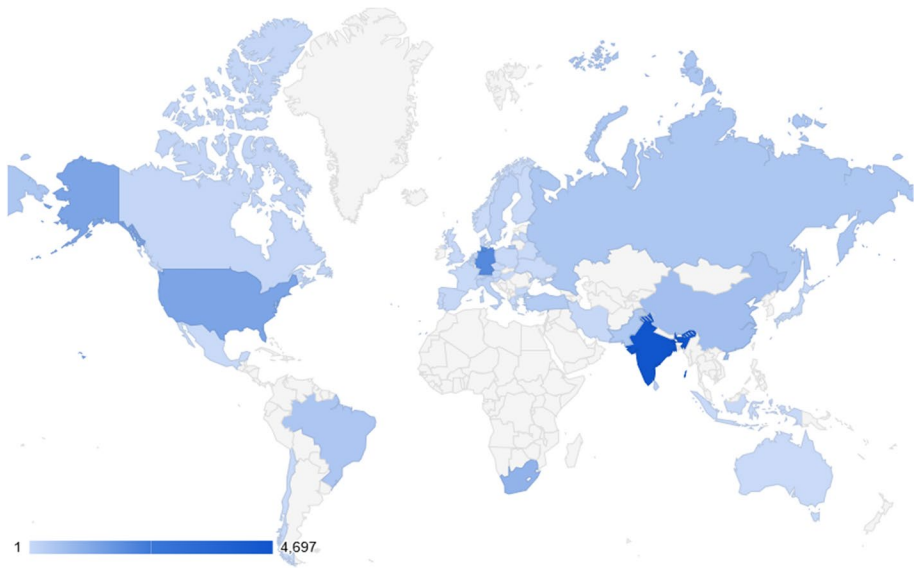


Fig. 3 Geographic representation of the participants from the selected studies

Table 6 Software engineering and related technical roles of selected studies

Role	Synonyms	Selected studies	#
Software development	Architects; back-end/front-end developers; computer programmers; computer information systems developers; computer scientists; core engineers; developers; engineers; maintainers; packagers; peripheral engineers; programmers; software coders; software developers; software development team members; software engineers; users of code; technical leaders; system architects	WB01, WB02, WB03, WB04, WB05, WB06, WB07, WB08, WB09, WB10, WB11, WB12, WB13, WB14, WB15, WB16, WB17, WB18, WB19, WB20, WB21, WB22, WB23, WB24, WB25, WB26, WB27, WB28, WB29, WB30, WB31, WB32, WB33, WB34, WB35, WB36, WB37, WB38, WB39, WB40, WB41, WB42, WB43, WB44	44
Product/project management	Product managers; product owners; program managers; project managers; scrum masters	WB01, WB05, WB09, WB12, WB13, WB17, WB25, WB26, WB33, WB34, WB35, WB39	12
Analysis	Analysts, business analysts; IS analysts; system analysts	WB01, WB02, WB05, WB09, WB12, WB13, WB16, WB24, WB25, WB26	10
Testing and QA	Bug reporters; QA; QA analysts; QA specialists; software testers; testers	WB01, WB02, WB09, WB17, WB25, WB26, WB28, WB36, WB41	9
People management	Department managers; general directors; managers; owners; presidents; supervisors; team leaders	WB02, WB04, WB05, WB07, WB12, WB14, WB18	7
Support and operations	Consultants; database; network; persons in charge of networks, CRM systems, and operation systems; researchers; technical consultants; translators	WB06; WB09, WB13, WB16, WB23, WB24, WB25	7
Design	Designers; graphic; UX/UI specialists; web designer	WB02, WB06, WB09, WB17, WB25, WB26	6
Other	Hobbyists; teachers; users of apps; other/unidentified	WB06, WB23, WB25, WB36, WB39	5

Table 7 Research methods of the selected studies

Research method	Selected studies	#
Cross-sectional	WB01, WB02, WB03, WB04, WB05, WB06, WB07, WB09, WB10, WB11, WB12, WB13, WB14, WB15, WB16, WB17, WB18, WB19, WB20, WB21, WB22, WB23, WB24, WB25, WB26, WB27, WB28, WB31, WB32, WB33, WB34, WB35, WB36, WB37, WB38, WB39, WB40, WB41, WB42, WB44	40
Longitudinal	WB08, WB30, WB43	3
Quasi-experimental	WB29	1

Table 8 Statistical methods of the selected studies

Statistical method	Selected studies	#
Structural equation modeling (SEM)	WB05, WB10, WB12, WB16, WB17, WB20, WB22, WB24, WB25, WB26, WB27, WB28, WB30, WB38, WB39, WB40, WB42, WB44	18
Regression analysis	WB03, WB04, WB06, WB07, WB11, WB13, WB14, WB15, WB18, WB19, WB23, WB32, WB33, WB34, WB36, WB37, WB43	17
Multilevel modeling	WB08, WB09, WB21, WB35, WB40	5
Correlation analysis	WB01, WB02, WB29, WB31	4

measures ($n = 5$), or used a combination of several approaches ($n = 4$). In some studies, researchers used measures that were neither indicated in the cited work nor available online ($n = 3$).

The most common measures of well-being are as follows: Utrecht Work Engagement Scale ($n = 4$) for integrated well-being; Satisfaction with Life Scale ($n = 4$), Index of Job Satisfaction ($n = 4$), and Job Diagnostic Survey ($n = 3$) for hedonic well-being; Balanced Measure of Psychological Needs ($n = 2$), Psychological Empowerment Instrument ($n = 2$), Method Control Scale ($n = 2$), and Work Design Questionnaire ($n = 2$) for eudaimonic well-being.

The majority of selected studies focused on hedonic ($n = 17$) or eudaimonic ($n = 13$) aspects of software engineers' well-being, whereas only a few studies looked at integrated or hedaimonic well-being ($n = 5$). A fair number of studies explored well-being multidimensionally by looking at several or multiple traditions of well-being ($n = 9$). The significant predictors and outcomes of increased (green arrows) and decreased (red arrows) well-being provided in Fig. 4 and 5.

4.3 A theory of Software Engineers' Well-Being

The results of the systematic literature review served as the input to create a multidimensional theory that ontologically defines and describes the predictors and outcomes of software engineers' well-being (Gruber 1993). In other words, the theory describes what well-being is, which factors predict well-being, and what are the outcomes of well-being. As such, the resulting theory qualifies as a Type III theory or theory of prediction, according to Gregor (2006).

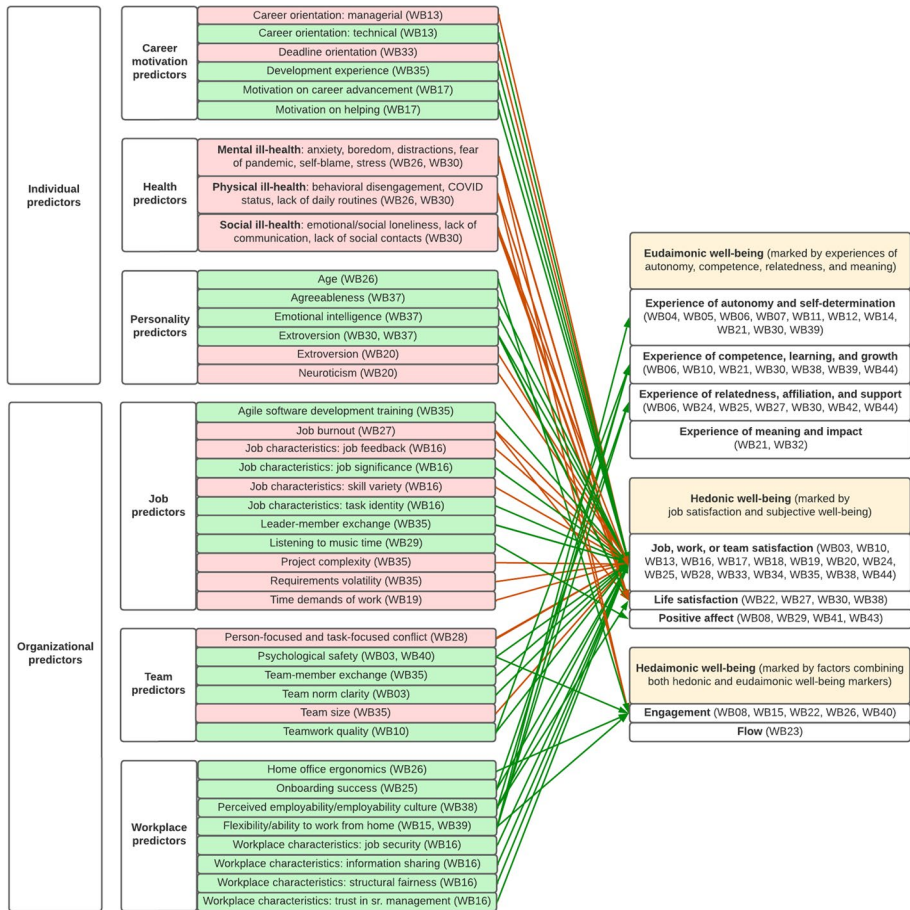


Fig. 4 Predictors of software engineers' well-being

The theory was developed using both inductive reasoning (i.e., from specific observations to general theory) and deductive reasoning (i.e., from general theory to specific conclusions), using the results of the literature review as input. As a result, well-being was conceptualized as a meta-construct of hedonic, eudaimonic, and hedaimonic or integrated well-being that affects the functioning of software engineers and their organizations, based on the combined results of the selected studies.

The theory and its constructs, propositions, explanations/ justifications, scope, and diagrammatic notations are further described according to the guidelines for theory building, proposed by Sjøberg et al. (2008). In contrast to the detailed summary of all statistically significant relationships that emerged from the systematic literature review (Fig. 4 and Fig. 5), the resulting theoretical model of well-being (Fig. 6) is represented through higher-level categories to emphasize the predictive power of the theory – the prime interest of many theorists (Gregor 2006).

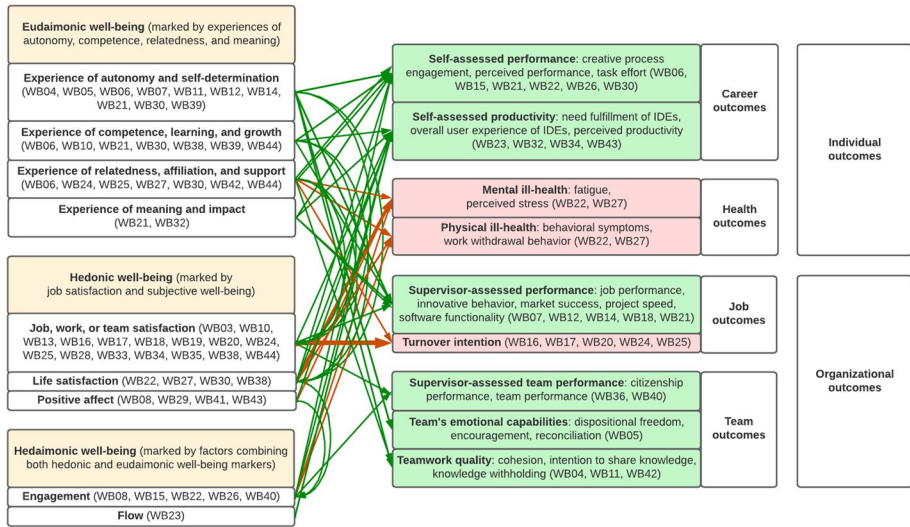


Fig. 5 Outcomes of software engineers' well-being

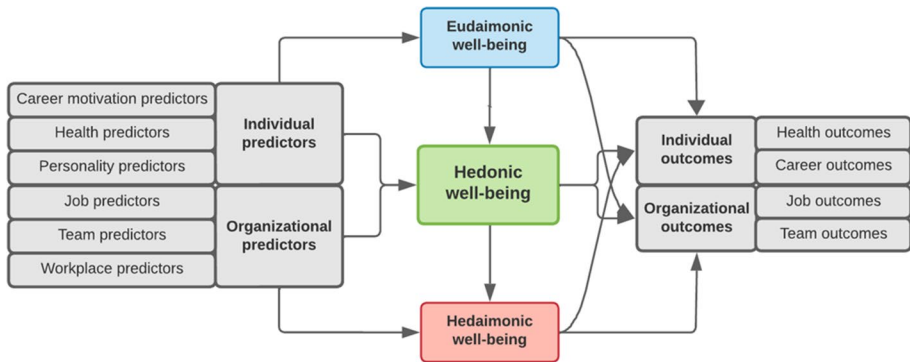


Fig. 6 Conceptual version of software engineers' well-being theoretical model

4.3.1 Constructs

The proposed theory is constructed by three types of elements as theory's building blocks: well-being, its predictors, and its outcomes. The theory divides well-being into three categories (see Fig. 6 in the middle): hedonic, eudaimonic, and integrated or hedaimonic well-being, which is a neologism that we coined to identify indicators of well-being that combine markers that share both hedonic and eudaimonic qualities. The predictors of well-being (see Fig. 6 on the left) are the factors that have been found to predict changes in well-being and are divided into individual factors (including career motivation, health, and personality) and organizational factors, including job, team and workplace predictors. The outcomes of well-being (see Fig. 6 on the right) denote the changes that well-being predicts on the two levels: individual outcomes (marked by health and career characteristics) and organizational outcomes, marked by job and team characteristics.

4.3.2 Propositions

The relationships between constructs (i.e., groups of deductively and inductively clustered variables) are based on statistical regressions extracted from the selected quantitative studies (see Fig. 4). Arrow means the direction of the regression. Color yellow signifies categories of well-being, whereas color green signifies that the coefficient of the regression is positive and color red means that it is negative. Information in brackets signifies the IDs of the selected well-being studies. The relationships can be summarized with the following three propositions:

Proposition 1: Software engineers' well-being is marked by three forms of well-being: hedonic, eudaimonic, and integrated or hedaimonic well-being.

Proposition 2: The well-being of software engineers is predicted by two groups of factors:

- a. individual factors (include software engineers' career motivation, health, and personality);
- b. organizational factors (include software engineers' job, team, and workplace characteristics).

Proposition 3: The software engineers' well-being predicts the functioning of software engineers (i.e., health and career outcomes) and their organizations (i.e., job and team outcomes).

All propositions of the theory are empirically and quantitatively testable. The exact relationships in the theory should be explored in future studies, using statistical analysis, such as regression analysis or structural equation modeling. Below each principle is described in greater detail.

Proposition 1: Software engineers well-being is marked by three forms of well-being: hedonic, eudaimonic, and integrated or hedaimonic well-being

All three forms of well-being are important when discussing software engineers' well-being.

Notably, satisfaction—the cognitive dimension of hedonic well-being—is considered the primary or core component of the model for at least two reasons. Satisfaction-related constructs (i.e., job and life satisfaction) were studied more frequently than positive affect, which is the emotional dimension of hedonic well-being. For this reason, satisfaction encompasses affect and therefore is considered the primary form of software engineers' hedonic well-being. Two, satisfaction was the most frequently studied category of software engineers' well-being in terms of its presence in regression analyses.

Although hedonic well-being is the primary element of the model, that does not mean other elements of well-being are less significant. On the contrary, the model hypothesizes that eudaimonic and hedaimonic well-being dimensions boost or strengthen the positive and well-established relationship between satisfaction and individual and organizational outcomes. The hypothesis is based on studies that found positive relationships between different forms of well-being (i.e., between hedonic and hedaimonic/eudaimonic). For instance, positive affect predicts work engagement (WB8). Similarly, life satisfaction

is predicted by several forms of eudaimonic well-being, that is, satisfaction of needs for autonomy (WB30, WB07), competence (WB30), and relatedness or relationships (WB30).

Proposition 2: The well-being of software engineers is predicted by individual and organizational factors

Individual factors, such as career motivation, health, and personality, are the software engineers' internal or psychological characteristics. Individual factors describe the qualities that software engineers bring to their team and organization. First, health describes three forms of ill-health: mental (e.g., anxiety stress), physical (e.g., behavioral disengagement, lack of daily routines), and social (e.g., emotional and social loneliness). Various forms of psychological dysfunction are negatively associated with hedonic well-being (i.e., life satisfaction) (WB30) and hedaimonic well-being (i.e., engagement marked by emotional well-being) (WB26). Second, career motivation describes a software engineers' career-related factors, such as development experience (WB35), deadline orientation (WB33), career orientation (WB13), eagerness to help colleague and advance one's career (WB17). Finally, personality encompasses various individual differences, such as age, emotional intelligence, and three traits of big five personality model: agreeableness extroversion, and neuroticism. It is important to note different studies report contradictory results regarding the relationship between extroversion and well-being: extroversion has a positive impact on life satisfaction but can have both a positive and negative impact on job satisfaction (see Fig. 4).

Organizational factors related to the job, team, and workplace are software engineers' external or sociotechnical characteristics associated with their work environment. Software engineers are exposed to these factors at their jobs, teams, and organizations. First, job predictors include several dimensions from the Job Diagnostic Survey: job feedback, job significance, skill variety, and task identity (Hackman and Oldham 1974). The remaining characteristics describe circumstances that improve well-being, such as leader-member exchange (WB35), agile software development training (WB35), and music listening (WB29), and those detrimental to well-being, including complex projects and volatile requirements (WB35), time demands of work (WB19), and job burnout (WB27). Second, team predictors describe software engineers' experiences associated with team dynamics and communication. These include team size (WB35), teamwork quality (WB10), psychological safety and team norm clarity (WB03), and two forms of conflict judgment, i.e., task-targeted and person-targeted (WB28). Finally, workplace predictors include flexible work arrangements with the ability to work from home (WB15, WB39), home office ergonomics (WB26), employability culture (WB38) and onboarding success (WB25), in addition to workplace characteristics encompassing job security, information sharing, trust in senior management, and structural fairness (WB16).

Proposition 3: The software engineers well-being predicts the functioning of software engineers (i.e., health and career outcomes) and their organizations (i.e., job, team, and workplace outcomes)

This model has two outcome groups: individual and organizational. Individual outcomes include career outcomes and health outcomes. Career outcomes describe subjective factors that are associated with the career success of software engineers, such as self-assessed performance and self-assessed productivity. Health outcomes are subjective factors that are associated with the physical and psychological functioning of software engineers. Some

of these dimensions are found in the definition of health provided by the World Health Organization⁶ (1948): “a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity”. However, our model distinguishes between individual health and individual well-being. In the model, individual health is defined as the absence of negative mental, physical, and social states, while individual well-being indicates the presence of positive states, that is, hedonic, eudaimonic, and integrated or hedaimonic well-being.

In addition to the success and functioning of individual software engineers, a technology organization is considered successful when software engineers contribute positively to certain job and team outcomes. Job outcomes are objective factors associated with the work-related success of software engineers. These include various indicators of supervisor-assessed performance, such as job performance, innovative behavior, and software functionality. Team outcomes are objective factors associated with the work-related success of software engineering teams. These include supervisor-assessed team performance, team’s emotional capabilities, and teamwork quality.

4.3.3 Scope of the Theory

Theories typically have a limited scope or conditions and circumstances, under which the theory propositions are valid (Sjøberg 2008; Stol and Fitzgerald 2015). Our theory is applicable to the well-being of professional software engineers operating in the social context of software engineering teams and technology organizations.

Our theory is constructed using empirical evidence from studying diverse software engineering roles, including software developers, analysts, product/project managers, testers, designers, people managers, and other technical professionals that were conceived as software engineers by other researchers. For this reason, the actual relationships between different predictors and outcomes may vary for different roles, genders, professional or individual identities, and organizational contexts. The latter is due to the fact that well-being may be influenced by the culture and values adopted in an organization, which are indirectly influenced also by the national contexts. Notably, unlike many other studies biased towards empirical research conducted in the Western societies, the evidence behind our theory comes from diverse nationalities. The evidence includes the following countries with the cumulative highest number of study participants from all around the world, including India ($n=4,697$); Germany ($n=1,918$); the US ($n=1,268$); South Africa ($n=924$); China ($n=637$); Brazil ($n=470$); Russia ($n=464$); Pakistan ($n=393$); Turkey ($n=348$); Sri Lanka ($n=232$); Italy ($n=189$); and Sweden ($n=165$).

Furthermore, contrary to many other theories of software engineers’ well-being, our theory goes beyond the professional and organizational realm, which is crucial for conceptualizing well-being as a multidimensional construct that includes physical, emotional, cognitive, and social aspects of individual and collective functioning (Dodge 2012; Lomas et al. 2014; Linton et al. 2016). Therefore, the insights gained from the theory can benefit both individual software engineers seeking to improve their well-being and work performance as well as the managers and leaders of software engineering teams and organizations working towards establishing a healthy culture and behaviors for their employees.

⁶ World Health Organization. (1948, April 7). Preamble to the Constitution of the World Health Organization (WHO). <https://www.who.int/about/who-we-are/constitution>

5 Discussion

Our systematic literature review results provide a fertile ground for a more nuanced and up-to-date conceptualization of software engineers' well-being (Singh and Junnarkar 2016). The developed multidimensional theory of software engineers' well-being incorporates the existing unidimensional models and theories.

In the following, we discuss the emerging model and how it relates to the existing relevant models.

5.1 Well-Being as a Complex Multi-Dimensional Construct

The interest in well-being as a core concept in organizational psychology in general and in various professional fields in particular is growing. This is evident in the increasing number of empirical studies on well-being of software engineers included in our literature review. Yet, the results of our literature review once again emphasize the elusive nature of well-being as a complex, multi-faceted construct that is hard to define and measure. Our mapping of the studied concepts confirms the prevalence of unidimensional constructs and a lack of multidimensional conceptualizations of well-being in empirical research in software engineering.

Our literature review is not the first review in the field. However, similarly to empirical studies, the majority of existing literature reviews focus on unidimensional states of well-being, such as emotions (Sánchez-Gordón and Colomo-Palacios 2019) and motivation (Beecham et al. 2008; França et al. 2011), or negative markers of well-being, such as stress (Maudgalya et al. 2006; Suárez and Vizcaíno 2023), time pressure (Kuuttila et al. 2020), and cognitive load (Gonçalves et al. 2021). In contrast, our review relied on and aggregated the results from a much more diverse set of quantitative studies that focused not only on software engineers' hedonic well-being (i.e., mainly satisfaction, less so affect) but also on eudaimonic well-being (i.e., experiences of autonomy, competence, relationships, and meaning and impact) and integrated or hedaimonic well-being (i.e., engagement and flow).

The most related to our work is the literature review by Salas-Vallina et al. (2018), who analyzed studies on knowledge workers' well-being from a multidimensional perspective, including hedonic well-being (e.g., job satisfaction and subjective well-being), eudaimonic well-being (e.g., psychological well-being and self-determination), and integrated well-being (i.e., engagement and flow), albeit the review was semi-systematic. Specifically, Salas-Vallina et al. (2018) analyzed a diverse number of categories and well-being indicators related to our own. For example, the authors discuss contextual performance marked by civic and proactive behaviors. We find contextual performance equally important, as it is related to eudaimonic well-being, i.e., one of the three categories of our model described as valued and growth-oriented living (Ryan and Martela 2016; Bartels et al. 2019; Unanue 2021).

Contrary to our review, Salas-Vallina et al. (2018) used somewhat different categorization of predictors and outcomes, which can be caused by the differences in the data used for abstraction and conceptualization. Another conceptual difference between Salas-Vallina et al. (2018) and our review is that the authors conceptualized psychological capital and commitment as well-being indicators. In our view, dimensions of psychological capital, such as optimism, efficacy, resilience, and hope, are better understood not as indicators of well-being but personal resources (in our model, individual factors) that contribute to the well-being and performance of employees (Luthans et al. 2007; Grover et al. 2018; Wu and

Nguyen 2019). Also, we argue that commitment (an indicator of well-being in Salas-Valina et al. 2018) is similar to continuance intention but the opposite of turnover intention (outcomes of well-being in our review).

5.2 Relationship with Related Models and Theories of Well-Being

In the following, we compare our work to existing models and theories in the software engineering literature (Beecham et al. 2008; Sharp et al. 2009; Graziotin 2016; Baltes and Diehl 2018; Storey et al. 2019; Alsunki et al. 2020). Notably, some authors of relevant models and theories do not necessarily refer to the studied concepts as hedonic, eudaimonic, and integrated/hedaimonic, or associate their studies with the boarder concept of well-being explicitly. Thus, in some cases, we adapted the models and theories according to our classification.

The results are summarized in Table 9.

5.2.1 Related Theories and Models of Hedonic Well-Being

Hedonic well-being emerged as the most extensively studied tradition of software engineers' well-being. Models and theories in software engineering focused on at least three markers of hedonic well-being: motivation (Beecham et al. 2008; Sharp et al. 2009), affect (Graziotin et al. 2015a, b; Graziotin 2016), and job satisfaction (Storey et al. 2019).

First, the Model of Motivation in Software Engineering was developed based on a systematic literature review by Beecham et al. (2008) and refined in the Model of Motivators, Outcomes, Characteristics and Context by Sharp et al. (2009). Both models focus on software engineers' motivators, and, like many other models of motivation, emphasize various organizational factors, such as autonomy, control, equity and career paths, and important outcomes, such as retention and productivity. The main advancement in these models is related to attention to the software engineers' personal characteristics (e.g., personality) and context (e.g., job characteristics).

Second, the Explanatory Theory of the Impact of Affects on Programming Performance proposed by Graziotin et al. (2015a, b) is based on deep observations and interviewing of two computer science students. The theory suggests that programmers' happiness is affected by the events they experience in their personal and professional lives, including private events (e.g., memories, reflections, and realizations) and public ones (e.g., situations observable from the third person). Depending on the situation, events can trigger either positive or negative affect (Graziotin et al. 2015a, b; Graziotin 2016). If programmers experience higher positive and lower negative affect, they focus better on task and project goals (associated with higher levels of flow), consequently increasing their performance. In other words, the model shows that hedonic well-being (positive affect) leads to hedaimonic well-being (flow). Noteworthy, the authors differentiate affect from satisfaction and do not consider affect nor flow as forms of well-being (Graziotin 2016).

Finally, the Theory of Software Developer Job Satisfaction and Perceived Productivity developed by Storey et al. (2019) is based on productivity-related literature and a survey. The theory promotes the social and technical factors, challenges, and work context variables that influence the relationship between job satisfaction and perceived productivity. Contrary to our review that considers job satisfaction an indicator of hedonic well-being or hedonic happiness, the authors argue that happiness is a broader concept than job satisfaction. That said, just because hedonic well-being is a broader category

Table 9 Mapping to related categories and variables from other theoretical models/theories (WB stands for well-being)

Categories and variables in our model/theory		Alsunki et al. (2020)	Storey et al. (2019)	Baltes & Diehl (2018)	Graziotin (2016)	Beecham et al. (2008), Sharp et al. (2009)
Hedonic WB	Job satisfaction	Extrinsic motives	Overall job satisfaction	N/A	N/A	Job satisfaction
	Subjective WB (life satisfaction and positive affect)	Extrinsic motives	N/A	N/A	Affect	N/A
	Experience of autonomy	Intrinsic motives	Autonomy	N/A	N/A	Autonomy; Need for independence
	Experience of competence	Competence of design; Intrinsic motives; Technical ability;	Personal technical skills	Knowledge and experience; Mental abilities/skills	N/A	Creativity; Growth-orientation; Need for feedback; Technical competence
Eudaimonic WB	Experience of relationships	Intrinsic motives	N/A	N/A	N/A	Need to be sociable
	Experience of meaning/impact	Intrinsic motives	Impactful work; Important contributor	Seeing usage	N/A	Making a contribution; Need to make a contribution
	Engagement	Developer engagement	N/A	N/A	N/A	Employee participation/involvement; External signs of motivation; Interest
	Flow	N/A	N/A	Motivation by work as challenge	Focus	Need for challenge; Need for variety/variety of work
Individual predictors	Health predictors	N/A	N/A	Mental abilities	Private events (non-work-related)	Stress
	Career motivation predictors	Attitude to the project; Intrinsic motives	N/A	Motivation by personal priorities; Motivation by helping others	Private/public events (non-work-related)	Achievement-orientation; Career stage; Managerial career path; Technical career path

Table 9 (continued)

Categories and variables in our model/theory						Beecham et al. (2008), Sharp et al. (2009)
	Personality predictors	N/A	Storey et al. (2019)	Baltes & Diehl (2018)	Graziotin (2016)	
Organizational predictors	Job predictors	Development environment; Project duration; Project complexity; Schedule/effort distribution; Stakeholder communication	N/A	Personality	Private events (non-work-related)	Individual differences; Introversion; Personality traits
			Can complete tasks; Can switch teams; Engineering system; Manager; Time to do work; Work	Freedom; Tool support	Private/public events (work-related)	Addressed development needs Empowerment and responsibility; Job type; Need for stability; Task identification/significance; Technically challenging work
Team predictors				Experimentation; Team structure	Private/public events (work-related)	Sense of belonging; Supportive relationships; Trust/respect
	Workplace predictors	Organizational culture; Top management support	Culture; Work-life balance; Work environment	Information exchange	Private/public events (work-related)	Appropriate working conditions; Career path; Equity; Good management; Work-life balance
Individual outcomes	Career outcomes	Developer productivity	Perceived productivity	Productivity	Programming performance	Absenteeism; External signs of demotivation; Productivity; Retention
	Health outcomes	N/A	N/A	N/A	N/A	N/A
Org outcomes	Job outcomes	N/A	N/A	Code quality; Quality	Programming performance	N/A
	Team outcomes	N/A	N/A	N/A	N/A	N/A

that encompasses characteristics other than job satisfaction, it does not mean that job satisfaction cannot be conceptualized as an indicator of hedonic well-being.

Evidently, there are many commonalities across the theories and models focusing on hedonic aspects of software engineers' well-being. First, most models study various predictors of hedonic well-being. Second, the majority of them focus on very narrow aspects of well-being. Finally, some models are generic and identify categories of confounding factors, while others are detailed and link specific factors with the impacted aspects of well-being. In our work, we depart from the mentioned models that focus on hedonic well-being in several ways (see Table 10 for a summary of differences and relevant practical implications).

First, our theory does not speculate about the exact relationships between individual and organizational factors. What it does suggest is that software engineers' well-being is affected by various individual and organizational factors, regardless of their exact relationships. For example, our model shows that higher levels of task or process conflict are associated with decreases in hedonic well-being (i.e., affect and satisfaction), which is confirmed by independent studies (Acuña et al. 2009; Kuriakose et al. 2019). However, the model does not speculate whether the conflict is due to a toxic team culture or a higher number of disagreeable people in the team. From a practical point of view, as long as any of the individual or organizational factors is addressed (e.g., whether it is by improving the communication in the team or encouraging individuals to seek professional support, such as counseling or coaching), the levels of software engineers' well-being should go up. Nevertheless, all theoretical models (including ours) are hypothetical, unless their effectiveness is verified in diverse organizational settings.

Table 10 Summary of key differences from related hedonic models and theories and relevant practical implications

#	Key differences from related hedonic models/theories	Practical implications
1	Well-being is affected by individual and organizational factors, regardless of their exact relationships	As long as individual or organizational factors are addressed (e.g., whether it is by improving the communication in the team or encouraging individuals to seek professional support, such as counseling or coaching), the levels of software engineers' well-being should go up
2	Our model assigns valence not to the individual or organizational factors but the state of well-being itself	Certain experiences may seem stressful or frustrating today (e.g., a very challenging task or project) but, upon reflection, turn out to be great sources of growth and meaning in the future (e.g., learning and growing from mistakes in a retrospective session) For example, a cyberattack on company's core infrastructure can induce stress and anxiety, but also act as a flow-inducing experience and sense of camaraderie
3	Both satisfaction and affect can be considered evaluative, theoretically inextricable concepts	When employees are surveyed about their feelings, they cannot but think about their affective states, which naturally changes or distorts their embodied phenomenological experience, especially when someone lacks self-awareness and emotional intelligence
4	If we want to understand software engineers, we have to understand what they think, feel, and want	A nuanced understanding of software engineers' experiences entails a combination of approaches ranging from quantitative research and sentiment analysis (e.g., judging states from text or work outputs) to human-to-human conversations: authentic, honest, and confidential

Second, our model does not specify whether certain antecedents or predictors are “positive” or “negative”, contrary to other models that talk about “demotivators” (Beecham et al. 2008; Sharp et al. 2009) or “challenges” (Storey et al. 2019). Instead, our model assigns valence not to the individual or organizational factors but the state of well-being itself, as suggested by Graziotin et al. (2015a, b), which, in our view, makes sense for two reasons.

1. It is very difficult to judge whether an experience is “positive” or “negative” using only cross-sectional data, that is, observations collected at one point in time. According to positive psychology 2.0 or the second wave of positive psychology, certain experiences may seem stressful or frustrating today (e.g., a very challenging task or project) but, upon reflection, turn out to be great sources of growth and meaning in the future (e.g., learning and growing from mistakes in a retrospective session) (Wong 2011; Ivztan et al. 2015). Unfortunately, longitudinal studies of software engineers’ well-being are expensive and, therefore, not very common.
2. Some events not only can seem but also feel positive and negative. For example, complex states, such as awe (i.e., wonder accompanied by fear) or *schadenfreude* (i.e., joy or pleasure open witnessing the misfortune of another person), can be experienced as both positive and negative affect (Berrios 2017a, b, 2018). For example, a cyberattack on company’s core infrastructure can induce stress and anxiety, but also act as a flow-inducing experience and induce a sense of camaraderie.

Third, in our conceptualization of hedonic well-being, we do not consider satisfaction and affect as theoretically different or distant concepts, as suggested by Graziotin et al. (2015a, b). On the one hand, both terms refer to slightly different psychological phenomena. For example, affect can be defined as “a neurophysiological state that is consciously accessible as a simple, non-reflective feeling that is an integral blend of hedonic (valence) and arousal values” (Russel 2003, p. 147), whereas satisfaction is defined as a “positive (or negative) evaluative judgment one makes about one’s job or job situation” (Weiss 2002, p. 175). In less technical terms, both terms constitute subjective experiences, whereby affect is more felt and “in our body” and satisfaction is more rationalized and “in our head.”

On the other hand, when subjects are asked to rate their emotions and moods in interviews or questionnaires, they automatically engage in various cognitive and neurolinguistic processes that constitute evaluation, including attention (e.g., focusing on the question), comprehension (e.g., understanding the question), memory (e.g., remembering situations when a particular affective concept was felt), awareness (e.g., becoming aware of one’s recalled affective state), and production (e.g., producing an answer based on the recall and awareness). For this reason, when employees are surveyed about their feelings, they cannot but think about their affective states, which naturally changes or distorts their embodied phenomenological experience, especially when someone lacks self-awareness and emotional intelligence. Unless affect is measured by physiological/behavioral measures or sentiment analysis (i.e., instruments that do not require conscious participation of the subjects), both satisfaction and affect can be considered evaluative, theoretically inextricable concepts. Therefore, our theory is better validated by the Theory of Software Developer Job Satisfaction, and Perceived Productivity developed by Storey, in which satisfaction is defined as cognitive and/or affective evaluation of one’s job (Wright and Cropanzano 2000). Their definition is similar to our conceptualization of subjective well-being marked by two evaluative components: cognitive

(i.e., satisfaction) and emotional (i.e., presence of positive affect and absence of negative affect).

Finally, the most important contribution of our model is the emphasis on the multidimensional nature of software engineers' well-being and the limitations of hedonic-only perspective. This is also supported by related research (Fagerholm and Munch 2012). According to Fagerholm and Munch (2012), if we want to understand software engineers' experience, we have to answer three questions: what software engineers think (i.e., cognition), feel (i.e., affect), and want (i.e., conation). Therefore, affect (i.e., the affective element of hedonic well-being) and cognition (i.e., the cognitive element of hedonic well-being) cannot address the whole picture of well-being, which is marked by various factors ranging from motivation and engagement to self-determination and meaning (Bakker, and Demerouti 2007; Oades and Dulagil 2016; Lysova et al. 2019; der Kinderen and Khapova 2020). Speaking practically, a nuanced understanding of software engineers' experiences entails a combination of approaches ranging from quantitative research and sentiment analysis (e.g., judging states from text or work outputs) to human-to-human conversations: authentic, honest, and confidential.

5.2.2 Related Theories and Models of Eudaimonic Well-Being

Eudaimonic well-being emerged as the second most prominent form of software engineers' well-being.

It can be argued that software engineers lead an eudaimonic life, especially in the Aristotelian sense of a "good life", when they behave responsibly (e.g., ensure systems security, avoid code plagiarism) and care about sustainability (e.g., think about the consequences of their actions on the users and the environment) (Sojer et al. 2014; Rahanu et al. 2019; Schieferdecker 2019; Nazir et al. 2020). However, it is likely that due to the complex nature of the job, software engineers are deemed to demonstrate unethical and irresponsible behavior, which begs the question of whether an Aristotelian-like eudaimonia can be achieved in the modern workplace (Stieb 2008). Nevertheless, we managed to find several models that focus on various indicators of eudaimonic well-being (Baltes and Diehl 2018; Malik et al. 2021), as defined by our theory: autonomy, competence, relationships, in addition to meaning and impact.

First, the Theory of Software Development Expertise developed by Baltes & Diehl (2018) is grounded in related research and empirical data from a mixed-methods survey with software developers. The theory focuses on software developers' expertise (the need for competence in our model), which is said to be affected by various individual factors (including motivation, and personality) and organizational factors (including task or work context). In other words, software engineers may experience eudaimonic well-being when their technical expertise and soft skills contribute to their job performance (Baltes and Diehl 2018). However, it is important to note that software engineers are more empowered to use their expertise and skills when the needs for autonomy or freedom are met (Baltes and Diehl 2018; Maturro et al. 2019).

Second, the Conceptual Framework of Psychological Empowerment developed as a research model by Malik et al. (2021) draws on related literature and an empirical survey of agile project managers and team members and conceptualizes agile practices as a source of intrinsic motivation that enables delivery of innovative outcomes. Their work demonstrated that various team-related factors, such as team communication (associated with team predictors in our model), positively impact eudaimonic well-being or psychological

empowerment marked by competence, impact, meaning, and self-determination. However, we excluded the model from Table 9 where we compare theoretical models and theories only.

As a final note, several models operationalize indicators of eudaimonic well-being as organizational factors that positively impact hedonic well-being (Beecham et al. 2008; Sharp et al. 2009). In our model, eudaimonic well-being also positively impacts hedonic well-being. Therefore, eudaimonic well-being can also be considered a predictor to hedonic well-being, as demonstrated by other quantitative studies (Diedericks and Rothmann 2014; Huta 2015; Joshanloo 2019). However, in our theory, we differentiate between the experiences of eudaimonic well-being (e.g., autonomy and competence) and techniques that contribute to eudaimonic well-being (e.g., less micromanagement and opportunities to demonstrate and grow skills and expertise), which, we argue, are better considered as predictors of well-being.

Practical recommendations In order to encourage software engineers to behave ethically and take personal responsibility for their behavior at work, avoid management practices that induce extrinsic motivation, such as corrective feedback, micromanagement, strict deadlines, and other forms of managerial control. Rather, prioritize practices that induce intrinsic motivation (marked by experiences of autonomy, competence, relationships, meaning and impact) and, consequently, increase job satisfaction.

5.2.3 Related Theories and Models of Integrated or Hedaimonic Well-Being

Integrated or hedaimonic well-being is a neologism that we created to define a research tradition that studies indicators of well-being containing both hedonic (e.g., I am enjoying what I do) and eudaimonic qualities (e.g., I am growing in what I do) (Tulip et al. 2020). In this tradition, we include flow and engagement (a broader category encompassing emotional well-being and work engagement). Our review demonstrated that flow and engagement were the most under-researched aspects of software engineers' well-being. That said, we managed to find several theoretical models engaging with concepts related to hedaimonic well-being.

For example, the Explanatory Theory of the Impact of Affects on Programming Performance states that hedonic well-being marked by positive affect increases performance when software engineers are focused (Graziotin et al. 2015a, b; Graziotin 2016). Graziotin (2016) connect focus to Mihaly Csikszentmihalyi's concept of flow, defining it as the "attention state of progressing and concentration" (p. 133). However, the authors do not specifically conceptualize flow as an indicator of well-being nor suggest that focus and flow are the same experiences.

Furthermore, the Framework of Software Developers Engagement Antecedents and Productivity developed by Alsunki et al. (2020), is based on the theory of workplace and human engagement, adopted for software engineers. The authors explain that software engineers' engagement is affected by four groups of variables: personal characteristics (associated with individual predictors in our model), group characteristics (associated with team predictors in our model), process/project (associated with job predictors), organizational factors (associated with work predictors in our model), and development environment (associated with eudaimonic well-being marked by competence). Similarly, Pratt et al. show that curiosity (an indicator of software engineers' personality) and goal clarity

(an indicator of software engineers' career motivation) are associated with higher levels of flow marked by enjoyment (Pratt et al. 2016).

Practical recommendations First, in order to improve focus and performance, make sure to provide software engineering with flow-inducing tasks that are not too boring or too stressful enough, that is, slightly above their experience and capabilities. Also, make sure to get software engineers curious and excited about the product or vision of the organization to increase enjoyment and passion at work.

5.3 Important Predictors of Software Engineers' Well-Being: Individual and Organizational Predictors

The software development literature contains a fair number of theoretical models that describe various predictors of software engineers' well-being (Maudgalya et al. 2006; Beecham et al. 2008; Graziotin 2016; Storey et al. 2019).

For example, the Explanatory Theory of the Impact of Affects on Programming Performance describes two types of predictors: private events (i.e., memories, reflections, and realizations that cannot be observed from the third person) and public events (i.e., situations that can be observed from the third person) (Graziotin et al. 2015a, b; Graziotin 2016). In our model, private events are related to individual factors that describe the psychology of software engineers, including their personality, career motivation, and health, whereas public events are related to organizational factors that describe software engineers' relationship with their job, team, and workplace environment. However, our theory does not suggest that individual factors are solely private (i.e., never observable) or organizational factors are solely public (i.e., always observable). For example, individual factors can be inferred from software engineers' output or observable behaviors (e.g., avoiding social situations, such as daily meetings or public speaking, can signify social anxiety, as demonstrated in Denizci Nazligul et al. 2019). Similarly, organizational factors can be inferred from software engineers' perceptions and attitudes to their job, team, and workplace.

Other models that discuss various predictors of software engineers' well-being include the Theory of Software Developer Job Satisfaction and Perceived Productivity (Storey et al. 2019) and the Model of Motivators, Outcomes, Characteristics and Context (Beecham et al. 2008; Sharp et al. 2009). All of these models share certain similarities with our model. First, they focus on satisfaction. Second, they discuss a list of predictors, although using other umbrella terms, such as challenges and sociotechnical factors (Storey et al. 2019) and intrinsic and extrinsic factors (Beecham et al. 2008; Sharp et al. 2009). Finally, the models include a variety of predictors similar to our individual factors (see Table 9 for similarities of factors between our model and other theoretical models and theories). However, as noted earlier, most of these models study a selected group of predictors in isolation rather than holistically.

Practical Recommendations First, while hiring, assess the personality traits and career motivation of the candidates. Specifically, pay attention to emotionally stable candidates who are motivated to help and advance their careers. Second, ensure that software engineers engage in tasks that have fulfilling and satisfying characteristics. These are tasks that can be performed from start to finish (i.e., task identity), allowing to make a positive impact within or outside the organization (i.e., job significance). Third, ensure satisfying

workplace characteristics, including fair pay and rewards, job security, and flexibility, as well as cultivate trusting relationships and transparent information sharing.

5.4 Important Outcomes of Software Engineers' Well-Being: Individual and Organizational Outcomes

In our theory, we provide a more holistic view of the outcomes of well-being classified into individual outcomes (career and health characteristics) and organizational outcomes (job and team characteristics). Many models and studies verify the positive relationship between various forms of well-being and different individual and organizational outcomes.

Software engineers' hedonic well-being (i.e., satisfaction and positive affect) contributes to better individual outcomes marked by career factors and organizational outcomes marked by job factors. In terms of career factors, these include higher levels of productivity (Storey et al. 2019) and goal achievement (Graziotin et al. 2015a, b; Graziotin 2016) and lower levels of absenteeism (Beecham et al. 2008; Sharp et al. 2009) and turnover intention (Sharp et al. 2009). Unfortunately, we could not find many studies that verify the predictive effect of software engineers' well-being on health, which we suggest as a prospective direction for future research. In terms of job factors, these include faster delivery time (Beecham et al. 2008; Sharp et al. 2009), quality (Sharp et al. 2009), profitability (França et al. 2011), and project success (Beecham et al. 2008; Sharp et al. 2009).

When it comes to eudaimonic well-being, the Theory of Software Development Expertise (Baltes and Diehl 2018) entails that software engineers are productive (e.g., commit frequently and resolve issues fast) and produce high-quality code (e.g., structured and maintainable) when their behavior is marked by general and task-specific knowledge and experience. As we already discussed, expertise is associated with eudaimonic well-being marked by competence. However, not only competence but many other aspects of eudaimonic well-being can positively impact performance, as demonstrated by multiple reviews and meta-analytic studies (der Kinderen and Khapova 2020; van den Broeck et al. 2021; Good et al. 2022). In terms of integrated or hedaimonic well-being, the Framework of Software Developers Engagement Antecedents and Productivity shows that engagement is associated with higher levels of productivity and performance (Alsunki et al. 2020).

Finally, while reviewing and comparing the selected studies with related studies, we noticed that not all models of software engineers' well-being differentiate between individual outcomes (i.e., subjective experiences that speak to the functioning and success of individual software engineers) and organizational outcomes (i.e., objective experiences that speak to the performance of software engineering teams and organizations). In our view, there are several reasons why, we believe, it is important to maintain a conceptual boundary between the two. First, it helps practitioners to decide when to use individual interventions like coaching and psychotherapy to target individual outcomes (e.g., career and health) as opposed to organizational interventions, such as team coaching and management/human resources consulting to target business outcomes. Second, from a software engineer's point of view, the boundary also helps software engineers to realize that achieving individual outcomes depends a lot on their personal actions and responsibility, whereas achieving organizational outcomes depends both on them and the leaders of their organizations.

Practical recommendations First, invest in your employees' future health and performance by providing opportunities for company-paid services, such mental health services

(e.g., psychotherapy, guided meditation subscriptions), sports and exercise (e.g., physiotherapy, gym subscriptions), and social physical activities (e.g., travelling or hiking together). Second, in order to improve team outcomes through eudaimonic well-being, help teams establish psychologically safe environments with clear norms, as well as rich and frequent communication. Also, support teams in resolving task and process conflicts.

6 Conclusions

In this paper, we presented our overview of research on well-being in software engineering based on a systematic literature review of carefully selected empirical studies employing quantitative analysis. We summarized the diverse measures of constructs and indicators of well-being, as well as its predictors and outcomes. Further, we proposed a Type III theory (a prediction-based theory) that describes what well-being is, how it occurs, and how it can be increased or decreased in practice.

6.1 Contributions to Research and Practice

Our review is the first literature review that (1) systematically evaluated the quantitative studies on software engineers' well-being in terms of their focus, demography, methodology, and quality; (2) provided a detailed, replicable, and quantitatively driven methodology for the interdisciplinary and multidimensional study of software engineers' well-being; (3) proposed a model and theory of the predictors and outcomes of software engineers well-being; and (4) confirmed the need for more quantitative studies in this field. The theory addresses the need for further conceptualization in the field of software engineering (Stol and Fitzgerald 2015) and provides a vocabulary and concept descriptions that can be reused by field researchers and harmonize future research studies focusing on well-being (Sjøberg et al. 2008; Stol and Fitzgerald 2015). Furthermore, the study attempted to introduce the goals and trends of positive psychology to software engineering while raising awareness of software engineers' suffering, which can be detrimental to their health and technology organizations.

The visual representation of the findings of our review in Fig. 4 and Fig. 5 can serve as a map of individual and organizational characteristics that predict well-being as well as its important outcomes, including higher levels of performance and productivity and lower levels of ill-health and turnover intention. Leaders can navigate this aggregated view to answer such questions as who to employ, what type of work to offer employees, how to coach and support teams, and how to organize the workplace.

6.2 Limitations

There are several limitations associated with the methodology of the systematic literature review.

1. The search strategy was based only on an automatic search to increase the study's replicability and reduce bias. At the same time, we might have omitted the most recent studies published in 2024 because we prioritized reproducibility of the search process over completeness by including articles published until the end of 2023 (full year).

2. Due to the high number of constructs and indicators of well-being, the review did not exhaust the number of terms that might be included in the search string.
3. Due to the study's focus on quantitative survey studies, qualitative or information extraction studies were not reviewed.
4. Because the study was initially conducted as a master's thesis, the data selection and quality assessment were conducted mostly by one reviewer, i.e., the paper's first author. However, the potential junior researcher bias in including/excluding the papers was assessed on a sample of a hundred papers, which was reviewed by the junior and a senior researcher and resulted in a full agreement.
5. Most selected studies used cross-sectional design and regression methods that do not offer information on causality between variables.

Notably, most of these limitations are caused by pragmatic reasons; otherwise, it would have been almost impossible to conduct the systematic literature review within reasonable time constraints. Therefore, replications of this study are recommended to diversify the search methods and search terms, and include studies of other methods.

The resulting theory of software engineers' well-being is also subject to several validity threats.

With respect to internal validity, we make no claims that the identified lists of predictors and antecedents of well-being of software engineers are exhaustive. In fact, the theory shall be viewed as a starting point that maps the specific measures of well-being from diverse studies in a multidimensional model. It identifies some studies and helps to evaluate the gaps that shall be further extended through future empirical studies. The resulting conceptualization of well-being (Fig. 6) is a holistic view of the theory. The identified categories of factors that explain well-being of software engineers overlap, extend and in some cases differ from a variety of existing models (see an overview in Table 9), because constructing theories is not a linear or sequential process with predictable outcome (Stol and Fitzgerald 2015). Also, different authors may use different grounds for conceptualization.

With respect to external validity, it is not clear whether the theory has a high predictive power in all organizational settings and for all software engineering roles. In fact, some primary studies included in our literature review suggest that context and even roles matter. For example, the authors of WB20 found conflicting evidence with respect to the impact of personality traits on well-being dependent on the role. Specifically, extroversion in system engineers has a positive impact on their job satisfaction, while in programmers the relationship between extroversion on job satisfaction is negative. For this reason, the impact of personality can differ from role to role and from organization to organization. Nevertheless, the building blocks of our theory are based on the self-reported experiences of 16,086 software engineering professionals from at least 42 countries. Therefore, we consider our theory more generalizable to the diverse population of software engineers than any existing theory or model developed using unidimensional measures of well-being and/or purely qualitative data.

6.3 Future Research Directions

We found that, despite the valuable work in the area of software engineers' well-being, the advances are fairly disparate and disjointed, as well as biased toward hedonic constructs, such as the absence of negative affectivity or presence of positive affectivity, as demonstrated by some of the reviews performed in software engineering research. With our

model and theory, we attempt to inform future research about the multidisciplinary nature of well-being and the interplay between its three core components: hedonic, eudaimonic, and integrated or hedaimonic well-being.

Notably, many existing studies that we reviewed do not specifically use well-being as the core phenomenon under study; similarly, different umbrella terms are used to refer to the predictors of well-being. One reason for this is the lack of reference vocabulary, which we hope we have addressed in our study. Another reason might be that studies of well-being in software engineering are conducted by software engineering researchers without deep knowledge and expertise in psychology. Our research team also faced challenges in developing a cross-disciplinary dialogue, which, we believe, was beneficial by contributing to mutual compassion and learning. We, therefore, support the emphasis on increasing collaboration between software engineering researchers and social scientists or psychologists as also pointed out by other scholars (Lenberg et al. 2015; Radant et al. 2016).

Finally, while our theory establishes the relationships between the indicators, predictors, and outcomes of well-being, further research is needed to understand the specific nuances of these relationships. Although we hypothesize that well-being could either mediate or moderate the relationship between various predictors and outcomes, it is also fair to assume that certain types of well-being could function as moderators (e.g., satisfaction), whereas others as mediators (e.g., eudaimonic and hedaimonic well-being). Further work shall focus on cross-disciplinary efforts to test the theory and its propositions in industrial and professional settings. Such efforts would not only deepen our understanding of the predictive and casual power of well-being in software engineering but also enrich and add more nuance to the current theory.

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Data availability The data that demonstrates all steps of the systematic literature review are openly available in an Online appendix – [available upon request]. The authors confirm that the data supporting the findings of this work will be available for the readers upon request as supplementary materials.

Declarations

Conflict of interest The authors have no financial or proprietary interests in any material discussed in this article.

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